MeOS Input Protocol

The purpose of the MeOS Input Protocol (MIP) is to provide an open standard for online punches and other competition data. The protocol is XML based.

MeOS implements a MIP Client and Melin Software provides a web server implementation (PHP + MySQL) under the Apache License[1] that can be used on your server as-is, or form a basis for a custom implementation.

Setting up the Web Server

The following procedure is a minimal setup of the MIP service on a web server.

File	Description	Remark
`functions.php`	Internal utility functions.	
`config.php`	MySQL and MIP configuration.	You must edit this file.
`input.php`	Invoked by MeOS or other MIP client. The client requests input from the web server.	
`enterpunch.php`	Simple web form where you enter the control number start number(s) for competitors that punched to control.	

1. Copy the files in the table above to your web server, which must support PHP.
2. Edit config.php to match the configuration of your web server's MySQL connection. (Note that this is most likely not the same MySQL server used by MeOS).
3. Start the Online Input service in MeOS. Enter the URL to your input `.php` script and an id number of the competition (the number should be unique for the competition; you enter this number in the enterpunch web form).
4. Enter punches using the enterpunch web from in your web browser, smart phone, etc.

Protocol Details

Data transfer to the web server is based on an XML format formally defined in mip.xsd. The MIP client send a http request to the server, with the following http headers

- `competition` set to a user defined competition id (number or string with numbers and letters a-z/A-Z).

[1] See http://www.apache.org/licenses/LICENSE-2.0

- `lastid` set to the id of the last item received from the server. The server is expected to respond with an XML file containing all data added posts since lastid (the response file also includes a new lastid, which the client uses for the next request).
- `pwd` An (optional) password as provided by the user.
- `response` is used instead of lastid when the connection is made in (requested, see below) response to input data. The value is the same number used in lastid for the initial request.

Informal description and some examples.

The XML response file (provided by the server) can include the following data

- Punches (control code, start number or card number, time).
- New or updated entries: name, class, club, card, starttime etc
- Cards (card number, start time (optional), finish time, punches)
- Status update (for example DNS)

When the MIP data (provided by the remote server) is used to make new entries, it may also request a MeOS response, by including the response section. MeOS will then make an additional connection to the server. This response includes an XML file containing the `MipStatusResponse`. The following response requests are supported

- `config` Includes the competition settings and classes that allows new entries (depends on settings in MeOS), including number of maps and course length. The data is sent using the MOP (See MeOS Online Protocol) format. This enables the service to automatically configure the classes that allows entry.
- `entrystatus` For each entry provided, a status (OK for a new accepted entry, ERROR for failed or denied entry, or UPDATEOK for an updated entry). The id and localId attributes identify the entry in the external system (Id) and in MeOS (localId) and a message provides a more information in case of a failed entry.
- `entries` Includes all current entries in the classes that allow new entries. The entries are provided using the MOP format. For additional requests, the output includes only the changes since the last request. This enables an application with direct entry to also show a complete start list its classes.

Notes

All times in the protocol is in tenths of a second, with some explicitly stated exceptions. The start time is given in tenths of a second after 00:00:00 (local time) on the (first) day of the event.

Example

The following is an example of a response to MeOS from the web server.

```
<MIPData xmlns="http://www.melin.nu/mip" lastid="3">
 <p code="33" card="12345" time="36070"/>
 <p code="50" sno="101" time="373250"/>
 <card number="12346">
  <start time="36000"/>
  <finish time="46000"/>
  <p code="33" time="37000"/>
  <p code="50" time="38010"/>
  <p code="100" time="39440"/>
 </card>
</MIPData>
```

From this data, we may conclude that a competitor with card 12345 punched control 33 at 01:00:07 and that a competitor with start number 101 punched control 50 at 10:22:05.

We also see that card number 12346 was read out, with a start time 01:00:00 and finish time 1000 seconds later (01:16:40), and the card contains punches 33, 50 and 100.

Entries

Version 2.0 also supports transferring entries. You can provide the name, club, card number, class, entry fee, and birth year (or date) of the competitor. However, only name and class are required.

You may also provide an id of the entry (which can later be used to update the entry). It is an integer between 1 and 10^9. Using a number outside that range results in undefined behavior.

In addition to the Id (which is the id from the external service), you can refer to existing entries by an a localId (which refers to the internal MeOS Id, also used in MOP) or by extId, which is the external Id (Used by IOF-XML).

Example

The following is an example of a response to MeOS from the web server.

```
<MIPData xmlns="http://www.melin.nu/mip" lastid="3">
 <entry id="1" classid="17" fee="70.0" paid="false">
   <name birthyear="1947">Snowman, Olof</name>
   <club>IF Thor</club>
   <card hired="true">12345</card>
 </entry>
 <entry id="2" classid="17" fee="70.0" paid="true">
   <name birthdate="2009-08-08">Queen, Elsa</name>
   <club>Arendal Wayfinders</club>
   <card hired="false">101010</card>
 </entry>
 <entry id="3" classid="17" fee="70.0" paid="false" notiming="true">
   <name birthdate="1976-03-13" sex="M" nationality="NOR">
      Frofunnen, Lost
   </name>
   <bib>101</bib>
   <phone>+45-1010234</phone>
```

```
    <rank>1501</rank>
    <dataA>1111</dataA>
    <text>Nice and easy!</text>
   </entry>
   <response type="entrystatus"/>
</MIPData>
```

Another possible response is the following, which is intended to set DNS (Did Not Start) status for a competitor with local MeOS Id 103. You can use either the two letter symbol NS (for not started) or the code 20 (See MOP for a complete list of status codes).

```
<MIPData xmlns="http://www.melin.nu/mip" lastid="4">
 <entry localId="103" classid="17">
   <name>Hallman, Rutger</name>
   <status>NS</status>
 </entry>
</MIPData>
```

Response Examples

To automatically configure an entry system, you could request a `config` response in the first response (when lastid = 0). And in addition, possibly `entries`, if you want to show a complete start list). The response could be:

```
<MIPData xmlns="http://www.melin.nu/mip" lastid="1">
 <response type="config"/>
</MIPData>
```

In response you would receive a `<MipStatusResponse>` with an embedded `<MOPComplete>` from the MeOS Online Protocol (MOP)

```
<MipStatusResponse>
  <MOPComplete>...</MOPComplete>
</MipStatusResponse>
```

When you work with new entries, you can request entrystatus to get the outcome of each entry, like is done in the example above. The response from MeOS in that case.

```
<MipStatusResponse>
  <EntryStatus>
  <Entry status="ERROR" id="1" localId="0" message="The class is full"/>
  <Entry status="OK" id="2" localId="23" message=""/>
  <Entry status="OK" id="3" localId="24" message=""/>
  </EntryStatus>
</MipStatusResponse>
```